Chandan Kumar Barman

VAPT / Penetration Testing Intern

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PROFESSIONAL SUMMARY

B.Tech Computer Science and Engineering student at KIIT University targeting VAPT and offensive-security internships. Practical lab experience with Nmap, Wireshark, Kali Linux, and web application security testing; familiar with Python scripting and MCP-based security-tool integration.

TECHNICAL SKILLS

Security Tools: Kali Linux, Nmap, Wireshark, Burp Suite, OWASP ZAP, SQLmap, Nuclei

Security Knowledge: OWASP Top 10, Web Application Penetration Testing Fundamentals, SQL Injection, XSS, Broken Access Control / IDOR, Authentication Vulnerabilities, SSRF, OSINT, Digital Forensics, Basic SOC Concepts

Networking & Web: TCP/IP, HTTP/HTTPS, DNS, REST APIs

Programming & Scripting: Python, Bash / Shell, PowerShell

Development & Integration: Git, GitHub, Custom Python MCP Server Configuration, AI-Assisted Security-Tool Orchestration

HANDS-ON TOOLS & KEY LEARNINGS

- **Kali Linux:** Used as my main lab environment for security practice. Learned to work comfortably in the Linux command line, manage security tooling, and set up isolated environments for safe testing.
- **Nmap:** Used in lab environments for host discovery and port/service scanning. Learned how open ports and service versions map out an attack surface, and how TCP/IP behavior shows up in scan results.
- **Wireshark:** Used to capture and inspect network traffic in lab setups. Learned to read packets and follow TCP, HTTP/HTTPS, and DNS conversations to understand how data actually moves across a network.
- **Burp Suite:** Used to intercept and modify HTTP/HTTPS requests against deliberately vulnerable web applications. Learned how requests, parameters, cookies, and sessions work, and how to test how an application handles unexpected input.
- **OWASP ZAP:** Used alongside manual testing for automated web scanning in lab applications. Learned that scanner output is only a starting point and must be manually validated before it can be called a real vulnerability.
- **SQLmap:** Used against deliberately vulnerable applications to explore SQL injection. Learned how injection arises from unsanitized input and why parameterized queries and input validation prevent it.
- **Nuclei:** Used for template-based vulnerability scanning, including within my MCP tool-integration project. Learned how scan templates work and why results need triage rather than being trusted blindly.
- **Python, Bash & PowerShell:** Wrote basic scripts and integration code independently to automate repetitive tasks and connect tools. Learned to read tool output programmatically and chain commands into repeatable workflows.
- **Model Context Protocol (MCP):** Configured a custom Python MCP server that exposes security tools to an AI client. Learned how tool definitions, execution, and result-passing work, and why controlled, authorized test environments matter when connecting AI to security tools.

VULNERABILITY CLASSES PRACTICED (LAB ENVIRONMENTS)

- **SQL Injection:** How unsanitized input reaches database queries, and how to test for and prevent it.
- **Cross-Site Scripting (XSS):** How unescaped output lets scripts run in a browser, and why output encoding and input handling matter.
- **Broken Access Control / IDOR:** How predictable identifiers and missing authorization checks expose other users' data.
- **Authentication Vulnerabilities:** How weak login, session, and credential-handling logic can be tested and where it breaks down.
- **SSRF:** How an application can be made to send server-side requests, and why internal-resource access is a risk.
- **OWASP Top 10, OSINT & Digital Forensics:** Used as the framework for organizing web-security practice, plus public-source information gathering and evidence analysis from CTF challenges.

WORK EXPERIENCE

Cyber Mentorship Intern May 2026 – June 2026

LaunchED Global

- Completed practical cybersecurity lab exercises as part of a Cyber Mentorship Internship.

PROJECTS

AI-Assisted Security Tool Orchestration Using MCP Python, MCP, Nmap, Burp Suite, Nuclei

- Configured a custom Python MCP server to integrate security tools into an AI-assisted security-testing workflow.
- Wrote integration code to support tool execution and retrieval of results.
- Tested the workflow against intentionally vulnerable applications in a controlled environment.
- Key learning: how AI clients call external tools through MCP, and the importance of scoping tool access to authorized lab targets.

Web Application Security Testing Lab Burp Suite, OWASP ZAP, SQLmap

- Practiced testing in deliberately vulnerable applications for SQL injection, XSS, broken access control/IDOR, authentication vulnerabilities, and SSRF.
- Used web security testing tools to investigate application behavior and explore common vulnerability classes.
- Key learning: distinguishing scanner output from validated vulnerabilities, and understanding both the attack technique and the defensive fix for each issue class.

EDUCATION

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, India Expected 2028

B.Tech in Computer Science and Engineering

ACHIEVEMENTS

CyberVault CTF Competition 2025

- Top 10 overall placement in cybersecurity challenges involving web security, digital forensics, and OSINT.

AVAILABILITY

Available for full-time internships; open to remote and on-site roles.